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Federated Learning for Traffic Flow Prediction With Synthetic Data Augmentation

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ABSTRACT Deep learning-based traffic prediction models require vast amounts of data to learn embedded spatial and temporal dependencies. The inherent privacy and commercial sensitivity of such data has encouraged a shift towards decentralised data-driven methods, such as Federated Learning (FL). Traditional machine learning models capture spatial and temporal relationships in centralised data. In reality, traffic data is likely distributed across separate data silos owned by multiple stakeholders. In this work, a cross-silo FL setting is motivated to facilitate stakeholder collaboration for optimal traffic flow prediction applications. This work introduces FedTPS, a novel framework which augments each client's local dataset with synthetic data generated by a federated diffusion-based model. The proposed framework is evaluated on three large-scale real-world datasets and assessed against various generative and spatio-temporal prediction models, including a novel prediction model we introduce which leverages Temporal and Graph Attention mechanisms to learn embedded Spatio-Temporal dependencies. Experimental results show that FedTPS outperforms several FL baselines with respect to global model performance.

INDEX TERMS Federated learning, traffic flow prediction, spatio-temporal modeling, generative models, synthetic data, data augmentation.

I. INTRODUCTION

MODERN Intelligent Transport Systems (ITS) have increasing levels of connectivity enabling data exchange between users and infrastructure; it is projected that by 2026, all vehicles on UK roads will have some degree of connectivity [1]. This data can be leveraged to develop traffic prediction models, which can enhance the safety and productivity of users, as well as deliver significant social, environmental, and economic benefits to a city's inhabitants. Deep learning-based models have demonstrated exceptional accuracy in their ability to model spatio-temporal relationships, and predict future traffic states [3], [4], [5], [6], [7].

The majority of previous works towards spatio-temporal prediction aim to train a model from a centralised perspective where the collective data is available on a single device. Centralising city-wide ITS data in real-world settings is a difficult task; ITS represent a partnership between stakeholders such as local government authorities, ride-sharing companies, and micro-mobility organisations. Sharing journey data has implications on user privacy, as well as commercial implications. Developing frameworks for decentralised data-driven approaches is critical for fostering stakeholder collaboration within an ITS, as they eliminate the need to centralise private and commercially sensitive information.

Federated Learning (FL) [8] provides a solution for training traffic prediction models collaboratively by decentralising the training process of a traffic prediction model. Such a framework is detailed further in Figure 1. Although recent works in FL demonstrate its capacity to train a global model which learns spatio-temporal dependencies across distributed data, these works do not directly address the issue of data heterogeneity [9], or instead focus on the client selection strategy or parameter aggregation function [10], [11], [12], [13], [14], [16].

To tackle the issue of data heterogeneity and imbalanced data partitions across clients, data sharing mechanisms for FL applications in highly heterogeneous settings were explored in [17] wherein a small partition of real data is distributed to clients to augment their local datasets. However, sharing real data between clients may violate privacy requirements.

Research within FL for image classification tasks have recently looked at training generative models to augment local

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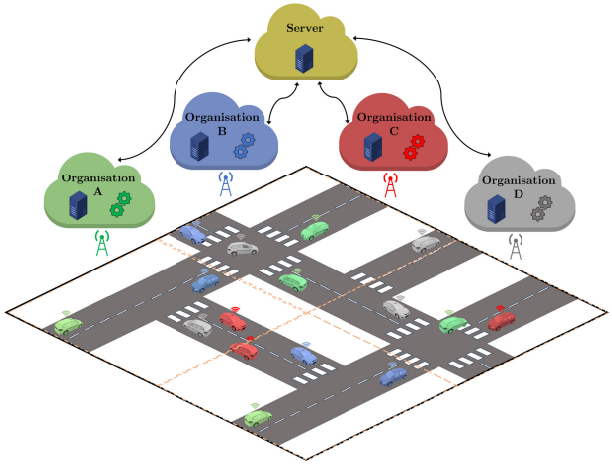


FIGURE 1. Federated framework for the traffic flow prediction task within an ITS. Each organisation collects data from their respective vehicle fleet, and through FL, collaboratively train a model to estimate traffic flow between regions.

client datasets without sharing data that would compromise data privacy [18], [19], [20]. For applications beyond image classification where data contains spatial and temporal dependencies and heterogeneity is principally due to quantity and feature-based skews, limited work exists on the benefits of augmenting FL client datasets using synthetic data.

In this work we aim to study the two interconnected tasks of distributed traffic flow generation and prediction. First, given partitioned trajectory data across multiple stakeholders within an ITS, how can we collaboratively train a generative model that produces synthetic traffic flow data capturing the global spatio-temporal characteristics without centralising or sharing raw data? Second, how can distributed clients collaboratively train a model to accurately predict future regional traffic flow when their local datasets exhibit heterogeneity?

Emerging research in the field of generative models have demonstrated the exceptional capacity for models to generate synthetic data that capture spatio-temporal interdependencies of traffic data [21]. Leveraging recent works in FL as well as in the field of generative models for spatio-temporal data generation, we present a framework for Federated Traffic Prediction with Synthetic data augmentation (FedTPS). This framework first trains a federated diffusion-based model for traffic flow data generation and generates a synthetic dataset developed from the learned global data distribution. A traffic flow prediction model is then trained through FL, wherein each client's local dataset has been augmented with a generated synthetic dataset.

In real-world implementations of traffic prediction models, the data owned by separate clients may contain inherent skews, contributing to data heterogeneity across client datasets. The proposed FedTPS framework directly inhibits the negative effects of data heterogeneity when training a global model through FL, while also increasing the amount of data in each client's local dataset. To the best of our knowledge, this work is the first to explore the benefits of augmenting the federated framework with synthetic data in traffic prediction applications.

The main contributions of our work are as follows:

- We present a novel traffic prediction model, the Graph Attention and Temporal Attention Unit (GATAU), which builds upon the existing spatio-temporal model, the Temporal Attention Unit (TAU) [6], to also leverage Graph Attention mechanisms.
- We propose a novel FL framework, FedTPS, which trains a federated generative model to produce a synthetic traffic flow dataset. This synthetic data is used to augment client datasets for traffic flow prediction, and improve the predictive performance of the global FL model.
- We assess the performance of several traffic flow prediction models trained under an FL setting. Our experiments are validated on two real-world datasets to analyse the impact of data augmentation on traffic flow prediction model performance, and demonstrate improved results relative to various FL frameworks designed to address data heterogeneity across data-silos.

II. RELATED WORK

A. FEDERATED LEARNING

The distributed nature of the data in an FL setting has a detrimental effect on the accuracy of the model relative to centralised training paradigms due to the fact that the distribution of data among clients is non-Independent and Identically Distributed (non-IID). In cross-silo FL settings, each client may represent a data center, owned by a particular organisation, which stores large amounts of data aggregated from many devices. FL has been extended to optimise the training of the global model [22], as well as to improve its applicability in non-IID settings [23]. Yet such methods focus on optimising the FL framework rather than address the data heterogeneity issues directly. FedDisco [24] addresses data distribution heterogeneity by incorporating a discrepancy metric into the aggregation weights. MOON [25] (Model-Contrastive Federated Learning) addresses the non-IID data challenge by introducing a contrastive learning objective at the model level, encouraging each client's local model to learn representations similar to the global model while diverging from previous iterations.

Existing data augmentation approaches for FL focus on image classification tasks, and typically rely on Generative Adversarial Networks (GANs) [18], [19], [20], or on sharing mean latent representations of client data [26]. Unlike image generation applications, traffic data generation requires modelling the dynamics of data over both spatial and temporal dimensions. Differentially-private representations of spatio-temporal data can be produced by altering real data [28], simulating traffic scenarios [29], or mixing partial aspects of the data [30]. These processes affect the spatio-temporal characteristics of the data itself, however.

Existing work on FL applied for traffic prediction implement novel client selection strategies to cluster clients with

similar spatial characteristics [10], [14]. Other such works look to adapt the aggregation procedure of the FL setting [11], [13], [16]. FedTDP [9] explores the data heterogeneity issue which may be found in real-world traffic data silos, yet does not propose a method for dealing with the inherent data heterogeneity across client data silos.

B. TRAFFIC FLOW MODELLING

To extract temporal dependencies, deep-learning based models typically employ either Recurrent Neural Networks (RNNs) based architectures [4], [12], temporal Convolutional Neural Networks (CNNs) which convolve around the time dimension of traffic flow data [3], [5], or temporal attention mechanisms to capture the evolution of the data over time [6].

Spatial relationships embedded within data can be captured through CNNs which convolve around the spatial dimension of data [6]. Traffic data can also be represented as graph networks, enabling Graph Convolutional Networks (GCNs) to capture spatial relationships within the data. GCN operations are dependent on the graph structure, which requires expert contextual knowledge to capture inter-region relationships. Graph Attention Networks (GATs) [32] implement a self-attention mechanism to learn the relationship between nodes.

To capture spatio-temporal dependencies within data, models such as STGCN [3] implement GCNs with temporal CNNs to capture spatio-temporal dependencies within data, while DCRNN [4] leverages GCNs with RNN-based models. GraphWaveNet [5] also utilises temporal CNNs and GCNs, however it learns an adjacency matrix for the graph structured data. The Temporal Attention Unit (TAU) [6] utilises CNNs and temporal Attention mechanisms to capture both spatial and temporal relationships within data.

Deep-learning based approaches have been explored for modelling and synthesising realistic and diverse spatio-temporal data [33], [34]. Conditional Generative Adversarial Networks (cGANs) [36] extend the standard GAN architecture by conditioning both the generator and discriminator on auxiliary variables (e.g. traffic density, or environmental context) to enhance control over the generation process. Similarly, Conditional Variational Autoencoders (cVAEs) [37] integrate conditional information into the latent space of the VAE framework. DiffTraj [21] is a conditional diffusion-based trajectory generation model which leverages Traj-UNet, a network based on the UNet architecture [38], to model the noise transitions at each time step in the diffusion process.

Recent works have addressed traffic data imputation and estimation in centralised settings [39], [40], yet such approaches cannot be directly applied to federated scenarios where data remains distributed and private. Furthermore, this work focuses on large-scale cross-silo FL settings where heterogeneity mainly stems from quantity and feature-based skews, and hence missing data is not the main challenge.

III. PROBLEM DEFINITION

Definition 1 (Traffic Flow Generation): The global dataset X_{glob} is partitioned into C clients such that $X_{\text{glob}} = (X_1 \cup X_2 \cup \ldots \cup X_C)$, where X_c contains client c's journey data (trajectories or origin-destination pairs). By discretising the city into N regions, we define $q_c^t \in \mathbb{R}^N$ as client c's regional inflow at time t. Given inflow sequences $Q_c = \{q_c^1, q_c^2, \ldots, q_c^m\}$, the objective is to learn a federated model G that generates synthetic inflows $Q_{\text{synth}} = \{q_{\text{synth}}^1, q_{\text{synth}}^2, \ldots, q_{\text{synth}}^m\}$ retaining the real global data's spatio-temporal characteristics.

Definition 2 (Traffic Flow Prediction): Given historical traffic flow $Q = (q^1, q^2, \ldots, q^T) \in \mathbb{R}^{N \times T}$, train a model to predict the inflow of all regions at future time step $T + \tau$, hence predict $q^{T+\tau}$. Based on the partitioned global data X_{glob}, client c can derive its regional traffic inflow data Q_c, and can also augment its own local dataset using the synthetic dataset Q_{synth}. The federated traffic flow prediction task is then to collaboratively train a global prediction model using the distributed client datasets with synthetic data augmentation, through FL.

IV. THEORETICAL BASIS

This section establishes the theoretical motivation for using synthetic data to reduce heterogeneity between clients in cross-silo ITS frameworks. Particularly, we elaborate on the theoretical basis for why synthetic data augmentation filters out client-specific patterns from the partitioning of city-wide traffic data.

A. CROSS-SILO DATA HETEROGENEITY IN TRAFFIC PREDICTION

This work pertains to the scenario of ITS stakeholders operating within a city, such as multiple ride-sharing companies (e.g., Uber, Lyft, Ola, local providers) or a combination of taxi fleets, micro-mobility services, and public transit authorities motivated to collaborate to deliver enhanced services for customers. Collaboration enables improved traffic flow prediction, route optimisation, and demand forecasting that benefits all participants. This naturally leads to a cross-silo federated learning setting where each organisation maintains control over their proprietary sensitive data while contributing to a shared predictive model.

In such a cross-silo federated learning for traffic prediction, each client organisation collects data over the same geographic regions and time periods, but with different traffic flow distributions. We model this heterogeneity using a mixture decomposition approach.

Let P_{global} represent the true underlying traffic flow distribution across the entire system. Each client c's observed distribution can be expressed as

$$P_c(x) = \alpha_c^{global} P_{global}(x) + \alpha_c^{sys} P_c^{sys}(x) \tag{1}$$

where $\alpha_c^{global} + \alpha_c^{sys} = 1$, $P_{global}(x)$ represents the true global traffic patterns, and $P_c^{sys}(x)$ captures systematic client-specific patterns.

In vertical cross-silo FL for spatio-temporal prediction, where stakeholders collect data over the same geographic span and time periods, α_c^{sys} is typically small relative to α_c^{global}.

B. FEDERATED DIFFUSION FOR GLOBAL PATTERN RECOVERY

The federated diffusion model in FedTPS learns an approximation $\hat{P}_{global}$ through the following optimisation

$$\hat{P}_{global} = \arg\min_{\theta} L(\theta) := \sum_{c=1}^{C} \frac{|Q_c|}{|Q_{\text{glob}}|} L_c(\theta), \tag{2}$$

where Q_{glob} is the global traffic flow dataset and $L_c(\theta)$ is the loss function for the diffusion model of client c (see Section V). Under federated training with sufficient rounds R_{global}, the federated diffusion model satisfies

$$\mathbb{E}[\hat{P}_{global}(x)] \to P_{global}(x) \tag{3}$$

as $|Q_{\text{glob}}|$ increases, and the client-specific component within $\bar{P}^{sys}$ vanishes; the diffusion model's denoising objective combined with federated aggregation creates a natural filtering effect of the client-specific component. During federated training, client-specific are averaged out across the federation, while consistent global patterns are reinforced.

C. SYNTHETIC DATA AUGMENTATION MECHANISM

When clients augment their datasets with synthetic data $Q_{\text{synth}} \sim \hat{P}_{global}$, we can analyse the effect on data heterogeneity through the mixture decomposition framework. Consider the augmented dataset where each client combines their original data with synthetic samples. The effective distribution that the client trains on becomes a mixture of their original distribution and the global distribution recovered by the federated diffusion model.

Let P_c^{aug} denote the effective distribution when client c augments their dataset with synthetic data from $\hat{P}_{global}$. If the augmented dataset contains a fraction $\lambda \in (0, 1)$ of synthetic samples, then $P_c^{aug}(x) = (1 - \lambda)P_c(x) + \lambda\hat{P}_{global}(x)$. This augmentation reduces data heterogeneity by reducing systematic client-specific patterns and reinforcing the global traffic patterns. In particular, since $\hat{P}_{global}$ contains no client-specific patterns, the weight of client-specific patterns in P_c^{aug} is $(1 - \lambda)\alpha_c^{sys} < \alpha_c^{sys}$. Similarly, the synthetic data introduces patterns learned from the entire federation, moving each client's distribution toward a more representative global distribution. When clients augment their datasets with samples from $\hat{P}_{global}$, they gain access to patterns that may be underrepresented in their local data but common across the federation. This regularisation effect reduces the variance in data distributions across clients, which is beneficial for federated learning convergence.

D. IMPLICATIONS FOR FEDERATED LEARNING PERFORMANCE

The reduction in systematic client-specific components and increased global pattern access provided by synthetic data augmentation directly addresses the core challenges of federated learning in non-IID settings. As established in the federated learning literature [23], [41], reduced client drift and improved distributional alignment lead to faster convergence and better global model performance. Our theoretical framework explains the specific mechanism by which synthetic data achieves these improvements in the context of traffic prediction.

V. METHODOLOGY

The proposed FedTPS framework first aims to train a generative model and generate a synthetic dataset from the global distribution of the journey data. The second step aims to train a traffic flow prediction model using each client's traffic flow data, and augment each client's data with the synthetic generated data.

A. FEDERATED GENERATIVE MODEL

A diffusion model is a probabilistic generative model that generates synthetic data through two sequential processes: 1) A series of forward processes which gradually injects noise into the data, and 2) The reverse of these processes, which learn to capture the original data distribution from a noisier version [42]. Recent works have applied diffusion-based models for high-fidelity data generation in various domains [43], [44], [45]. The forward process is defined as a Markov chain with F Gaussian transitions that map the real data $x_0 \sim g(x_0)$ to x_F, a latent variable with the same dimensionality as x_0, according to

$$g(x_1, \ldots, x_F|x_0) = \prod_{f=1}^{F} g(x_f|x_{f-1}), \tag{4}$$

$$g(x_f|x_{f-1}) = \mathcal{N}(x_f; \sqrt{1-\beta_f}x_{f-1}, \beta_f\mathbf{I}), \tag{5}$$

where β_f denotes the variable variance which controls the Gaussian noise at step f. Importantly, the forward process allows sampling x_f in closed form from $x_f = \sqrt{\bar{\alpha}_f}x_0 + \sqrt{1-\bar{\alpha}_f}\epsilon$, where $\epsilon \sim N(0, \mathbf{I})$ and $\bar{\alpha}_f = \prod_{i=1}^{f}(1-\beta_i)$. The reverse process can be parameterised as a Markov chain with learned Gaussian transitions

$$p_\theta(x_0, \ldots, x_{f-1}|x_F) = p(x_F)\prod_{f=1}^{F} p_\theta(x_{f-1}|x_f), \tag{6}$$

$$p_\theta(x_{f-1}|x_f) = \mathcal{N}(x_{f-1}; \mu_\theta(x_f, f), \sigma_\theta(x_f, f)^2\mathbf{I}), \tag{7}$$

where $x_F \sim \mathcal{N}(0, 1)$, and where $\mu_\theta(x_f, f)$ and $\sigma_\theta(x_f, f)$ are the mean and variance, parameterised by θ. The objective of the diffusion model is to minimise the error between the Gaussian noise ϵ and the predicted noise level $\epsilon_\theta(x_f, f)$, over its training set. For a data sample x_0, the client loss L_c is given by

$$\mathcal{L}_c(\theta) := \mathbb{E}_{x_0 \in X_c, \epsilon, f}\left[||\epsilon - \epsilon_\theta(x_f, f)||^2\right]. \tag{8}$$

The UNet architecture [38] has proved highly effective in generative modelling tasks; the fusion of multiple scales allows a diffusion model built on UNet to robustly navigate

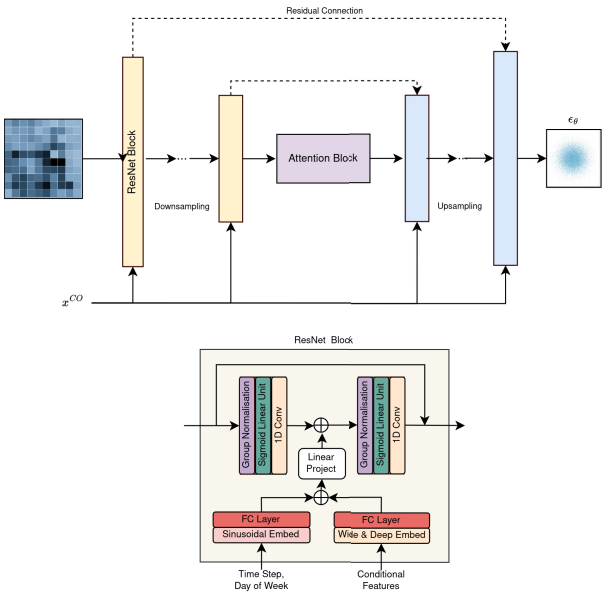


FIGURE 2. Architecture of TFDiff, based on the U-Net architecture.

noise removal in each denoising step of the diffusion process. UNet employs a contracting encoder, which applies convolutional operations to progressively reduce spatial resolution and extract salient features, followed by an expanding decoder that upsamples these encoded representations to restore the original spatial dimensions. Skip connections link corresponding encoder and decoder layers to retain spatial information from multiple scales that may otherwise be lost during downsampling; as the decoder upsamples the encoded input, the skip connections provide finer-scale information from the earlier encoded layers.

For the trajectory generation task, Zhu et al. [21] introduce an attention-driven transition module, and demonstrate its ability to bolster the spatio-temporal modelling capacity of diffusion-based methods. Akin to their DiffTraj model, our Traffic Flow Diffusion model (TFDiff) utilises the UNet architecture, with an intermediary attention block. Figure 2 illustrates the architecture of the UNet with intermediary attention modules for the task of traffic flow data generation.

TFDiff is conditioned on information regarding the time interval, the day of the week, the maximum traffic flow at any region within the time interval, and the region index with the maximum traffic flow.

For each dataset and client configuration, we train separate TFDiff models to capture city-specific traffic patterns.

1) FEDERATED FRAMEWORK

In a scenario where the global trajectory dataset is segmented across data silos from multiple organisations, a generative model cannot be trained under a traditional framework since the data is not centralised.

Under a federated setting, the updated parameters from each client's local training on the global model are communicated to a central server which then aggregates the results to update the global model. FedAvg aggregates the parameters based on the number of data samples in a client's partition. The objective of the federated TFDiff model is to minimise the loss of the model over the cross-client datasets, as shown in Equation 2.

The process of training a federated Diff model is illustrated in step one of Figure 3, and is detailed further in Algorithm 1. Once the global TFDiff model has been trained, it is shared with all clients, who then use it to generate the synthetic datasets to augment their respective local datasets.

Algorithm 1 Training the TFDiff Model Through FL Using FedAvg

Server initialises global model θ
Server selects all clients for participation, $c \in C$
for global round, $r_{\text{glob}} \in \{1, \ldots, R_{\text{glob}}\}$ **do**
  for each client $c \in C$ in parallel **do**
    Distribute global model to clients, $\theta \rightarrow c$
    for local round $r_{loc} \in (1, \ldots, R_{loc})$ **do**
      Sample $x_0 \sim g(X_c)$
      Sample $f \sim \text{Uniform}\{1, \ldots, F\}$
      Sample $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
      Local updates $\theta^c_{r_{loc}+1} \leftarrow \theta^c_{r_{loc}} - \eta \nabla_\theta \mathcal{L}_c$
    end for
    Global updates $\theta_{r_{\text{glob}}+1} \leftarrow \sum_{c=1}^{C} \frac{|Q_c|}{|Q_{\text{glob}}|} \theta^c_{r_{\text{glob}}}$
  end for
end for

The conditional data derived from each client's private dataset, as well as the distributed global TFDiff model, are used by each client to generate synthetic datasets for augmentation. The effect of varying the amount of synthetic data generated is studied through empirical analysis in Section VI.

TFDiff operates on spatially and temporally aggregated traffic flow data, which provides several privacy-enhancing properties. The diffusion model never accesses raw trajectory data, as vehicle movements are aggregated into regional counts over 30-minute windows. Furthermore, the discrete aggregated data is also obtained from a varying set of vehicles. Both of these points significantly increases the difficulty of individual trajectory reconstruction compared to raw GPS data, and act as privacy enhancing mechanisms.

B. FEDERATED TRAFFIC FLOW PREDICTION TASK

Using each client's local individual trajectory dataset X_c, the regional inflow dataset, Q_c, can be constructed, where $Q_c \in \mathbb{R}^{T \times N}$, T is the number of regional traffic flow interval samples, and N is the number of regions within the city. Each client can augment their local dataset with the synthetic regional inflow dataset Q_{synth} generated from each client using the TFDiff model for inference. Step two in Figure 3 illustrates the structure of the FedTPS framework for the traffic prediction task.

It should be noted that FedTPS operates at the data level through synthetic data augmentation, and hence this method is

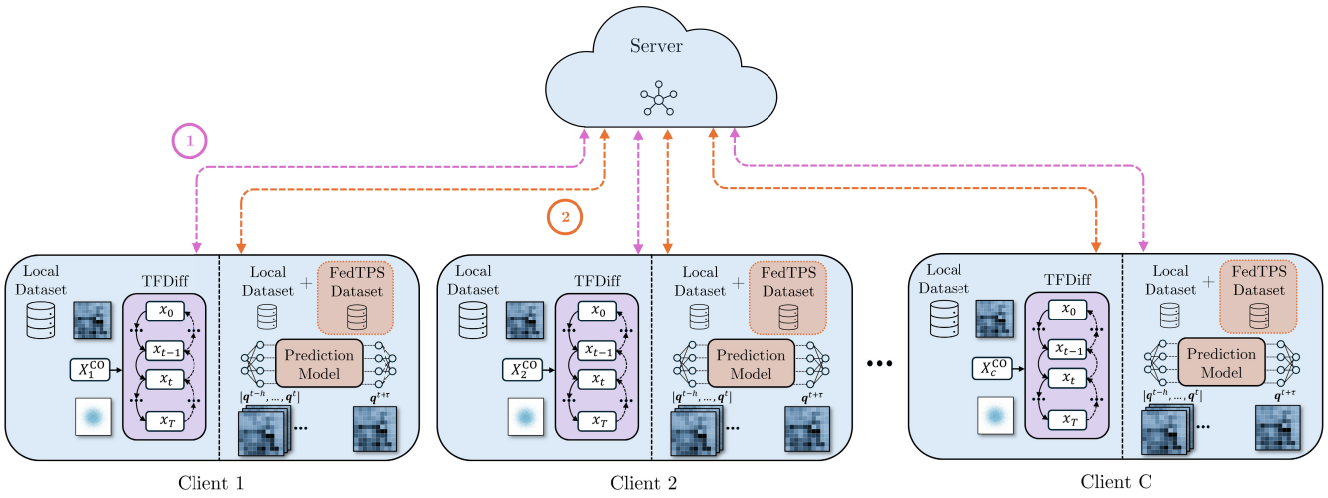


FIGURE 3. FedTPS Framework for training a federated generative model and subsequent traffic flow prediction model. The first stage, shown as the purple dashed line, trains the federated diffusion model to develop the FedTPS synthetic dataset. The second stage, shown as the orange dashed line, trains the federated traffic flow prediction model with the client's local dataset, as well as the synthetic data. X_c^{CO} represents client c's conditional observations required for the TFDiff model.

complementary to any federated mechanism which focuses on either client selection, or on the aggregation of client models.

The local objective of a traffic flow prediction model is denoted by

$$f_c(\omega) = \frac{1}{|Q_c|} \sum_t (\mathcal{L}(\mathcal{F}_c(\omega; q_{t-h:t}), q_{t+\tau}) \tag{9}$$

where the model is parameterised by ω, h denotes the number of previous traffic flow data samples fed to the model, and τ is the number of time steps in the future for which we will be predicting traffic flow. The global objective is therefore

$$\text{argmin} f(\omega) := \sum_{c=1}^{C} \frac{|Q_c|}{|Q_{\text{glob}}|} f_c(\omega) \tag{10}$$

where $|Q_c|$ and $|Q_{\text{glob}}|$ denote the number of regional inflow data samples in the dataset of client c, and in the global set of all clients, respectively.

C. PROPOSED TRAFFIC FLOW PREDICTION MODEL

The Temporal Attention Unit was introduced by Tan et al [6] and poses temporal attention as a combination of statical attention, which occurs within a data sample, and dynamical attention, which occurs between the frames of the data samples. The statical attention can be captured by performing depth-wise convolution, followed by a dilated depth-wise convolution, and finally a 2D convolution. The depth-wise convolution layers learn separate kernel filters for each of the channels in the encoded representation of the input. The dynamical attention is composed of a 2D average pooling layer, followed by a fully-connected layer. The final temporal attention is a result of the product between the statical and dynamical attention components.

Algorithm 2 Training a Global Traffic Flow Prediction Model Through FedTPS, Using FedAvg as the Base Aggregation Method for the Federated Method

Server distributes trained TFDiff model θ to all clients
Server initialises global prediction model ω
Server selects all clients for participation, $c \in C$
for each client $c \in C$ in parallel **do**
    Client c generates synthetic data X^c_{synth} using θ and local conditional data X^c_{CO}
    Client c augments local dataset: $Q_c^{aug} = Q_c \cup Q_c^{\text{synth}}$
end for
for global round, $r_{\text{glob}} \in \{1, \ldots, R_{\text{glob}}\}$ **do**
    for each client $c \in C$ in parallel **do**
        Server sends global model to client: $\omega \rightarrow c$
        for local round $r_{loc} \in (1, \ldots, R_{loc})$ **do**
            Local updates on augmented data: $f^c_{r_{loc}+1}(\omega) \leftarrow f^c_{r_{loc}}(\omega) - \eta \nabla_\omega \mathcal{L}_c(Q_c^{aug})$
        end for
        Client sends updated model to server: $f^c_{r_{\text{glob}}}(\omega) \rightarrow$ Server
    end for
    Server aggregates: $\omega_{r_{\text{glob}}+1} \leftarrow \sum_{c=1}^{C} \frac{|Q_c|}{|Q_{\text{glob}}|} f^c_{r_{\text{glob}}}(\omega)$
end for

An Encoder-Decoder architecture model architecture is presented in this work, wherein the Encoder and Decoder represent stacked 2D convolutional layers that encode a spatial representation of the data. To augment this Encoder-Decoder architecture for the task of traffic prediction, we can model the regional traffic flow data as a sequence of graphs, and employ the Graph Attention model [32] with multi-head attention, to learn multi-channel relationships between regions in the city.

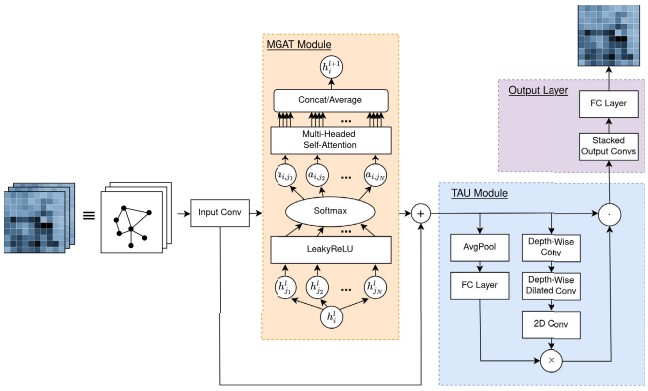


FIGURE 4. Architecture of the proposed GATAU model.

Thus, we propose the Graph Attention with Temporal Attention Unit (GATAU). Figure 4 illustrates the GATAU module in further detail.

Our input data $Q \in \mathbb{R}^{N \times T}$, where N is the number of nodes and T is the length of the input sequence, is first processed by a 2D convolutional layer to map the input into a higher dimension feature space. This output is fed into the Multi-Head Graph Attention (MGAT) module. Within the MGAT, attention coefficients are calculated which give a measure of the importance of relationships between nodes. The input into MGAT are node features, $h = \{\vec{h}_1, \vec{h}_2, \ldots, \vec{h}_N\} \in \mathbb{R}^{T \times F}$ where F is the input feature size. Attention coefficients for layer l are calculated by

$$e_{ij}^{l} = \text{LeakyReLU}\left(\vec{a}^{l^T}\left[W^l\vec{h}_i^l || W^l\vec{h}_j^l\right]\right) \tag{11}$$

where $W \in \mathbb{R}^{F' \times F}$ is a learnable weight matrix, F is the feature dimension of the input, F' is the output feature dimension which can be of different cardinality to F, and a is a function which computes the attention score from the concatenation of linearly transformed embedding of neighbouring nodes. We further apply a LeakyReLU activation for the attention coefficients. A fully connected linear single layer is used to represent function a. Then we normalise the attention scores across all connections for a given node through

$$a_{ij}^{l} = \frac{\exp(e_{ij}^{l})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik}^{l})} \tag{12}$$

where $\mathcal{N}_i$ denotes the node connections of node i. The output features for every node are calculated as a linear combination of the normalised attention coefficients from Equation 12 to give

$$h_i^{l+1} = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} W^k \vec{h}_j\right) \tag{13}$$

where σ is an ELU activation function. To enhance the stability of the model during training, a multi-head mechanism of attention is utilised for the graph attention module, wherein each attention head trains its own parameters. As per the original work which averages the multi-head outputs at the final layer [32], our model also averages the multi-head outputs after the single-layer MGAT module. The output from the MGAT module is of shape $[N, T, F']$.

A residual connection is used to fuse the output of the initial convolution layer with the output from the MGAT module. The fused hidden layer is then reshaped into $[N, F', H, W]$ format to be processed by the TAU module, where H and W are the dimensions of the rectangular grid of the city. Our output layer is composed of a convolutional layer, and fully connected layer, which reduce the dimensionality of our processed input's feature space to match our target shape, which is $Y \in \mathbb{R}^N$.

TABLE 1. Dataset characteristics for traffic flow prediction experiments.

Dataset	Chengdu	Xi'an	New York
Orders	5,807,001	3,410,420	8,914,743
Latitudes	(30.65, 30.73)	(34.21, 34.28)	(40.70, 40.85)
Longitudes	(104.04, 104.13)	(108.91, 109.00)	(-74.02, 73.93)
Time	01/11/2016-30/11/2016		01/01/2016-31/01/2016

VI. EXPERIMENTS

A. DATASET

The experiments in this study utilise the ride-sharing trajectory dataset published by Didi Chuxing GAIA Initiative spanning the cities of Chengdu and Xi'an in China during November 2016, as well as the New York origin and destination dataset published by the Taxi & Limousine Commission of New York, United States during January 2016. The different characteristics of these datasets provide a diverse study to evaluate FedTPS's ability to enhance federated spatio-temporal prediction models under different settings. Table 1 contains details of the datasets.

B. EVALUATION METRICS

To evaluate the performance of our generative and predictive models, we employ the following metrics. Let q_r^t denote the actual regional traffic inflow for region r at time t, $\hat{q}_r^t$ denote the predicted/generated value, and $\bar{q}$ denote the mean regional inflow across all regions and time steps in the dataset.

For traffic flow generation, the metrics employed assess distributional similarity between generated and real traffic flow:

- **Normalised Mean Absolute Difference (nMAD):** Measures the mean absolute difference between synthetic and real regional inflow distributions, normalised by mean regional inflow:

$$\text{nMAD} = \frac{1}{\bar{q}} \cdot \frac{1}{|T| \cdot |R|} \sum_{t \in T} \sum_{r \in R} |\hat{q}_r^t - q_r^t| \tag{14}$$

  where R is the set of all regions and T is the set of time steps. This provides a measure of how well the generated data matches the real data distribution.

- **Structural Similarity Index (SSI):** Measures the structural similarity between spatial patterns in generated and real data.
- **Temporal Correlation (TC):** Measures the temporal correlation between generated and real traffic flow sequences.

For traffic flow prediction, to evaluate prediction accuracy against ground truth future traffic flows, we use:

- **Normalised Mean Absolute Error (nMAE):** The MAE between predictions and ground truth, normalised by mean regional inflow:

$$\text{nMAE} = \frac{1}{\bar{q}} \cdot \frac{1}{|T| \cdot |R|} \sum_{t \in T} \sum_{r \in R} |q_r^t - \hat{q}_r^t| \quad (15)$$

- **Mean Absolute Percentage Error (MAPE):** The mean absolute percentage error between predictions and ground truth, quoted as a percentage (units in %):

$$\text{MAPE} = \frac{100}{|T| \cdot |R|} \sum_{t \in T} \sum_{r \in R} \left| \frac{q_r^t - \hat{q}_r^t}{q_r^t} \right| \quad (16)$$

  Specifically for the MAPE metric calculation, we only consider region-time pairs with real traffic inflows above one to avoid disproportionate bias.

C. DATA PRE-PROCESSING

The trajectory and order datasets are partitioned into subsets based on the number of clients (organisations) emulated in the FL setting.

The partitioned trajectories are processed into regional vehicle inflow format by dividing Chengdu and Xi'an into 10×10 regions, while the taxi trip records are processed into regional inflow format by dividing New York into 8×12 regions. Inflows over each region are aggregated over 30 minute intervals. For the data generation task, the partitioned traffic flow datasets, as well as the conditional dataset, are used to train TFDiff to generate synthetic traffic flow data possessing the same underlying spatio-temporal characteristics as the real data with some generational diversity. Then, FedTPS augments each client's traffic flow data with the generated synthetic data. Sequences of traffic flow data over previous 6 hours are used by the traffic prediction models to predict the regional inflow 3 hours in the future. The regional inflow data are divided into training, evaluation, and testing datasets, with the first 70% used for training, 10% for validation, and the final 20% for testing.

The properties of the partitions detailed in Table 2 suggest that as the number of clients in a setting increases, data heterogeneity also increases.

D. DESCRIPTION OF EXPERIMENTS

1) **Traffic Flow Data Generation:** This experiment details the performance of the data generation models as a function of data heterogeneity and number of clients. This is important as it demonstrates the generative capability of the traffic flow generation models in federated implementations with greater data heterogeneity.

TABLE 2. Metrics assessing the average similarity between the partitioned datasets representing each client's local dataset for each FL setting. $\bar{q}$ denotes the mean regional traffic inflow per time interval for each client case.

City	Clients	# Orders per Client Partition	$\bar{q}$	SSI	TC
Chengdu	2	(2904846, 2902155)	105.03	0.97	0.93
	4	(1452404, 1452464, 1453202, 1448931)	52.51	0.94	0.88
	6	(969168, 965928, 969817, 968308, 969601, 964179)	35.01	0.91	0.85
	8	(488639, 485265, 473309, 473086, 472199, 471045, 472957, 474078)	26.26	0.88	0.81
Xi'an	2	(1635623, 1686826)	72.91	0.92	0.94
	4	(841194, 844346, 841598, 845189)	36.45	0.85	0.89
	6	(562312, 560911, 562395, 561577, 561008, 564124)	24.30	0.80	0.85
	8	(421940, 419310, 422529, 421863, 421219, 420376, 421667, 423423)	18.23	0.76	0.81
New York	2	(4457538, 4457205)	24.86	0.98	0.62
	4	(2228350, 2228802, 2229188, 2228403)	12.35	0.95	0.59
	6	(1485446, 1485766, 1485975, 1485127, 1486117, 1486312)	8.29	0.93	0.56
	8	(1113494, 1114404, 1114471, 1114021, 1114856, 1114398, 1114717, 1114382)	6.11	0.92	0.52

2) **Effects of Varying Synthetic Data Volume:** This experiment details the performance enhancement of FedTPS as a function of the amount of synthetic data used.
3) **Traffic Flow Prediction** The purpose of this experiment is to assess how traffic flow prediction models perform over a range of different client cases under different FL frameworks, including FedTPS. This study is significant, as it emphasises how different federated mechanisms affect the performance of a global traffic flow prediction model, and highlights the benefits of a federated approach relative to traffic flow prediction models trained independently on separate data partitions.
4) **Pre-Training with Synthetic Data:** This study explores the effects of pre-training the global model in an FL framework using synthetic data. This study highlights pre-training as a method for training traffic flow prediction models in scenarios with limited communication or training rounds.

5) **Computational Overhead Analysis:** This study evaluates the computational and communication costs associated with the baseline spatio-temporal prediction models, as well as the TFDiff model. This analysis demonstrates the practical feasibility of FedTPS in real-world cross-silo federated learning deployments.
6) **Varying Global and Local FL Training:** The purpose of this study is to explore the benefits of FedTPS when the ratio of local training rounds to global parameter aggregation rounds is increased. Increasing the number of local training rounds per global aggregation can lead to the local training processes of each client to drift from one another, which can detrimentally impact the parameter aggregation process.
7) **Scalability Considerations:** This study looks at relative improvements between FedTPS when implemented with FedAvg aggregation relative to standalone FedAvg as client configurations vary from 2 to 12 clients.

E. BASELINES

For the traffic flow data generation task we use 100 global training rounds where we aggregate the updated client parameters after three local training rounds, and utilise the Adam optimiser with a learning rate of 0.0001 and a batch size of 32. We implement various baseline models including TFDiff.

1) **CVAE:** A conditional VAE with a fully connected encoder and decoder, each containing three hidden layers, and featuring a latent space dimension 20.
2) **CGAN:** A conditional GAN with three fully connected hidden layers in both the generator and discriminator, and taking a latent vector of size 100.
3) **TFDiff:** A conditional diffusion model. The encoder and decoder blocks in the UNet architecture feature multiple 1D-CNN-based stacked residual network blocks (Resnet block) with a kernel size of 3. An intermediate block, which is an attentive layer between two Resnet blocks with kernel size 1, is implemented between the encoder and decoder blocks.

For the traffic flow prediction task we use 80 global training rounds where we aggregate the updated client parameters after one local training round, and utilise the Adam optimiser with a learning rate of 0.001 and a batch size of 32. The baselines are implemented with the following settings:

1) **GRU:** A three-layer GRU model [7] with hidden dimensions of 32 is implemented.
2) **STGCN:** The STGCN model [3] using a distance based adjacency matrix.
3) **DCRNN:** The DCRNN model [4] using a distance based adjacency matrix.
4) **GWNET:** The GWNET model [5] using a self-adaptive adjacency matrix, where the graph connects regions within 4 kilometres.
5) **TAU:** The TAU model [6].
6) **GATAU:** The input convolutional layer is composed of two sequential 2D convolution layers with a hidden size 16. The TAU units have the same model parameters as in the original work [6]. Our output layer is composed of two 2D convolution layers at output with hidden size 16, and a kernel size (1,1), one final linear layer to match dimensions to the output dimension. For the Graph Attention layer, the graph connects regions within 4 kilometres.

To assess our proposed FL framework FedTPS, we implement a variety of different training frameworks.

1) **Non-FL Setting:** Each client trains traffic prediction model on its own partitioned dataset, and the best performing model is recorded.
2) **FedAvg:** FedAvg [8] uses a weighted averaging mechanism based on client dataset sizes, to update the global model based on individual client contributions.
3) **FedOpt:** FedOpt [22] builds on FedAvg by implementing server-side adaptive optimisers such as Adam, as well as server-side momentum. For the server, we use a learning rate of 0.01, a momentum parameter of 0.9, and a second moment parameter of 0.99.
4) **FedProx:** FedProx [23] utilises a penalty term, μ, to inhibit deviations of clients from the global model. We use a proximal term μ of 0.001.
5) **MOON:** MOON [25] augments local training with a model-contrastive loss that aligns local model representations with the global model while diverging from the previous local model. We set $\mu = 1.0$ and $\tau = 0.5$.
6) **FedDisco:** FedDisco [24] extends FedAvg by incorporating distribution discrepancy alongside dataset size when computing aggregation weights. We use hyperparameters $a = 0.5$ and $b = 0.1$.
7) **FedTPS:** FedTPS is the proposed framework that generates synthetic data to augment client datasets. As a data augmentation method, FedTPS is orthogonal to aggregation strategies; we study combinations with FedAvg and FedDisco below.

The code for this project is made available at https://github.com/FerOroC/FedTPS_Traffic_Prediction.

VII. RESULTS

A. TRAFFIC FLOW DATA GENERATION

Table 3 details the performance of various generative models based on three accuracy metrics which compare the similarity between the generated data, and the real data. The generative ability of the TFDiff diffusion model to generate traffic data with the same spatio-temporal characteristics as the real data is illustrated by the much lower nMAD, and the very high structure similarity and temporal correlation metrics. Although the CGAN and CVAE models achieved high accuracy scores on the SSI, the high nMAD and low TC scores demonstrate a high discrepancy between the real and synthetic data. This is due to the fact that cVAE and cGAN primarily capture spatial distribution, which may result in insufficient representation of the temporal dependencies inherent in traffic flow data.

TABLE 3. Comparison of generated regional inflow data against real regional inflow data for various centralised generative models.

City	Model	nMAD	SSI	TC
Chengdu	CGAN	0.53	0.86	0.33
	CVAE	0.55	0.93	0.53
	TFDiff	0.14	0.97	0.92
Xi'an	CGAN	1.02	0.78	0.22
	CVAE	1.40	0.88	0.29
	TFDiff	0.26	0.93	0.91
New York	CGAN	2.39	0.50	0.11
	CVAE	1.43	0.87	0.28
	TFDiff	0.22	0.98	0.60

TABLE 4. Comparison of generated regional inflow data against real regional inflow data for the TFDiff model trained through FL, with varying client cases.

City	Clients	nMAD	MAPE	SSI	TC
Chengdu	2	0.13	16.67	0.97	0.91
	4	0.18	21.21	0.93	0.86
	6	0.20	22.26	0.91	0.82
	8	0.23	23.77	0.88	0.79
Xi'an	2	0.31	31.25	0.94	0.88
	4	0.33	29.04	0.91	0.85
	6	0.38	29.90	0.88	0.81
	8	0.43	31.47	0.86	0.78
New York	2	0.25	22.42	0.96	0.57
	4	0.30	22.23	0.95	0.54
	6	0.33	22.73	0.93	0.51
	8	0.38	24.07	0.92	0.48

For the TFDiff model trained under federated settings with a varying number of clients, Table 4 shows the similarity between generated samples and real traffic data based on four metrics, averaged over the regional inflow data sequences. Some deviation is expected due to the generational diversity of the DiffTraj model, yet the high SSI and TC metrics demonstrate that the spatio-temporal characteristics of the original data were modelled accurately by the trajectory generation model. In general, increasing the number of clients in the FL setting results in a larger deviation from the real traffic data for the samples generated, as evidenced by the decreasing values of SSI and Temporal Correlation, as well as increasing nMAD and MAPE metrics.

Table 2 shows that the data heterogeneity in the partitioned datasets increases when there are more clients in the federated setting. This can cause the global model to struggle generalising across the multiple data silos, which leads to model performance degradation with higher client numbers, and slower global model convergence. Yet the samples generated throughout all client cases demonstrate high degrees of similarity with the real data, as the structural similarity index and temporal correlation among regions remain close to 1, and the nMAD remains low throughout the different cases.

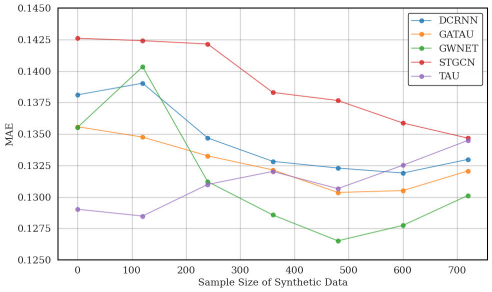


FIGURE 5. Effects of varying synthetic data sample size on average test nMAE for the city of Xi'an.

B. EFFECTS OF VARYING SAMPLE SIZE OF SYNTHETIC DATA

Figure 5 illustrates the test nMAE metric as a function of the varying synthetic dataset size used within the FedTPS framework. When no synthetic data is used, the federated framework is identical to FedAvg. For all models bar the TAU model, adding synthetic data of sample size 240 or higher decreases the nMAE. Exceeding the sample size of 600 seems to give diminishing returns with respect to this metric. A sample size of 480 (10 days) is selected to showcase FedTPS' performance in the ensuing experiments.

C. TRAFFIC FLOW PREDICTION

Table 5 presents the results of the traffic prediction models, trained under different FL paradigms with varying client numbers, using nMAE and MAPE metrics.

FedTPS is implemented using the weighted aggregation mechanisms of both FedAvg and FedDisco to demonstrate its performance enhancement capability across different FL methods. Incorporating FedTPS improves model performance for both aggregation strategies relative to their standalone versions, and both implementations consistently outperform other FL frameworks on the Chengdu and Xi'an datasets.

We attribute FedTPS's improved performance to the increased amount of data available to the clients during training, as well as the data heterogeneity reduction stemming from the shared synthetic dataset. Through synthetic data augmentation, we ensure that each client's dataset is more representative of the global data distribution, and increase the amount of information available to each client.

Comparing FedAvg, FedOpt, and FedProx, there is no consistently superior method across all datasets and client configurations. This is consistent with wider research on cross-silo FL with real-world data, which found that none of these FL methods consistently outperformed FedAvg across all settings [46].

MOON and FedDisco show modest improvements over FedAvg across Chengdu and Xi'an configurations. The benefits of different FL approaches appear to be both

TABLE 5. Comparison of various model performances for traffic prediction under varying client numbers and FL methods.

City	Clients	FL Method	GRU		STGCN		DCRNN		GWNET		TAU		GATAU	
			nMAE	MAPE	nMAE	MAPE	nMAE	MAPE	nMAE	MAPE	nMAE	MAPE	nMAE	MAPE
Chengdu	2	Non-FL	0.2385	35.54	0.1601	32.98	0.1528	27.56	0.1437	24.57	0.1145	20.66	0.1205	22.74
		FedAvg	0.2187	24.03	0.1249	23.21	0.1174	20.56	0.1145	19.86	0.1054	17.85	0.1142	19.29
		FedOpt	0.2190	23.90	0.1186	21.88	0.1152	20.26	0.1149	20.19	0.1043	17.91	0.1144	19.01
		FedProx	0.2167	25.02	0.1215	23.07	0.1214	21.10	0.1186	21.28	0.1122	18.61	0.1129	18.88
		MOON	0.2246	23.51	0.1196	21.00	0.1168	18.80	0.1153	19.83	0.1054	17.08	0.1147	18.25
		FedDisco	0.2288	23.45	0.1197	20.77	0.1171	19.09	0.1119	18.25	0.1073	17.21	0.1145	18.23
		FedTPS + FedAvg	**0.2143**	22.91	**0.1122**	19.88	**0.1065**	17.96	**0.1080**	18.69	**0.1019**	17.11	0.1090	18.02
		FedTPS + FedDisco	0.2221	**22.12**	0.1155	**19.39**	0.1068	**17.46**	0.1088	**17.40**	0.1022	**16.44**	**0.1074**	**16.81**
	4	Non-FL	0.2512	27.65	0.1758	30.47	0.1750	26.91	0.1651	25.01	0.1487	22.66	0.1472	21.59
		FedAvg	0.2331	24.31	0.1426	23.72	0.1394	22.67	0.1362	21.73	0.1285	20.62	0.1369	21.42
		FedOpt	0.2329	24.33	0.1442	24.26	0.1426	22.94	0.1384	22.02	0.1289	20.44	0.1373	21.22
		FedProx	0.2325	24.72	0.1455	24.75	0.1430	22.93	0.1386	22.38	0.1341	21.36	0.1373	21.21
		MOON	0.2421	23.77	0.1432	21.94	0.1419	21.59	0.1381	20.90	0.1330	20.16	0.1397	20.37
		FedDisco	0.2414	23.90	0.1438	22.18	0.1431	21.39	0.1390	20.91	0.1322	20.15	0.1402	20.44
		FedTPS + FedAvg	**0.2272**	23.28	**0.1360**	22.62	**0.1316**	21.22	**0.1282**	20.39	0.1335	20.89	**0.1299**	20.38
		FedTPS + FedDisco	0.2308	**23.06**	0.1378	**21.54**	0.1352	**20.60**	0.1284	**19.91**	**0.1306**	**19.55**	0.1326	**20.01**
	6	Non-FL	0.2614	32.31	0.1962	33.88	0.1954	31.03	0.1877	28.70	0.1760	26.66	0.1660	26.01
		FedAvg	0.2462	25.54	0.1588	24.58	0.1602	23.59	0.1557	23.02	**0.1485**	21.86	0.1571	22.78
		FedOpt	0.2459	25.35	0.1602	24.99	0.1614	24.01	0.1554	23.02	0.1488	21.91	0.1562	22.48
		FedProx	0.2442	25.65	0.1614	25.45	0.1620	24.23	0.1557	23.39	0.1528	22.57	0.1557	22.66
		MOON	0.2539	25.12	0.1602	23.61	0.1606	22.95	0.1574	22.50	0.1513	21.41	0.1581	21.94
		FedDisco	0.2519	25.07	0.1603	23.58	0.1613	22.80	0.1582	22.49	0.1521	21.49	0.1574	21.97
		FedTPS + FedAvg	**0.2402**	24.57	**0.1551**	24.09	**0.1525**	22.62	**0.1502**	22.15	0.1542	22.69	**0.1502**	22.05
		FedTPS + FedDisco	0.2485	**24.41**	0.1561	**22.99**	0.1535	**21.91**	0.1518	**21.55**	0.1492	**21.14**	0.1539	**21.33**
	8	Non-FL	0.2738	28.08	0.2091	31.36	0.2114	29.24	0.2057	28.62	0.2038	27.52	0.1836	25.78
		FedAvg	0.2590	27.17	0.1813	26.73	0.1824	26.02	0.1748	24.98	0.1691	24.00	0.1771	24.45
		FedOpt	0.2586	27.23	0.1805	27.14	0.1813	25.62	0.1756	25.64	**0.1687**	23.79	0.1760	24.47
		FedProx	0.2597	27.18	0.1855	27.81	0.1832	25.99	0.1763	25.34	0.1706	24.23	0.1782	24.74
		MOON	0.2645	26.71	0.1817	25.62	0.1811	24.59	0.1782	24.06	0.1732	23.21	0.1805	23.82
		FedDisco	0.2675	26.84	0.1814	25.22	0.1836	24.75	0.1799	24.64	0.1726	**23.20**	0.1801	23.47
		FedTPS + FedAvg	**0.2556**	26.83	**0.1767**	26.50	**0.1741**	24.60	**0.1691**	24.39	0.1801	24.87	**0.1725**	23.79
		FedTPS + FedDisco	0.2633	**26.27**	0.1783	**24.86**	0.1781	**24.18**	0.1737	**23.52**	0.1731	23.42	0.1753	**23.35**
New York	2	Non-FL	0.6620	37.17	0.4572	35.91	0.3973	33.54	0.3964	33.60	0.2770	22.84	0.2800	23.76
		FedAvg	0.5889	28.18	0.3255	25.41	0.2778	22.85	0.2875	22.80	0.2503	19.88	0.2555	20.35
		FedOpt	0.5867	28.00	0.3185	24.62	0.2813	22.78	0.2778	23.05	0.2481	19.83	0.2643	20.51
		FedProx	0.5937	28.25	0.3308	25.60	0.2844	22.80	0.2896	23.22	0.2542	20.06	0.2713	21.43
		MOON	0.5898	28.48	0.3224	25.42	0.2853	23.87	0.2824	23.67	0.2402	19.58	0.2604	21.53
		FedDisco	0.5961	28.20	0.3211	25.50	0.2890	23.69	0.2880	23.87	**0.2362**	**19.07**	0.2614	21.53
		FedTPS + FedAvg	**0.5863**	**27.54**	**0.3124**	**24.41**	**0.2752**	**22.00**	**0.2612**	**21.47**	0.2546	20.64	**0.2542**	**20.07**
		FedTPS + FedDisco	0.5917	28.11	0.3187	25.35	0.2858	23.39	0.2690	22.88	0.2396	19.61	0.2608	21.51
	4	Non-FL	0.6735	37.06	0.5073	34.55	0.4155	30.89	0.4277	32.03	0.3307	24.03	0.3125	23.33
		FedAvg	0.6051	30.51	0.3523	25.60	0.3143	22.70	0.3186	22.95	0.2943	21.23	0.2969	21.52
		FedOpt	0.6086	30.57	0.3394	24.93	0.3177	22.73	0.3091	22.74	0.2969	21.43	0.2969	21.35
		FedProx	0.6121	30.56	0.3549	25.49	0.3203	23.02	0.3143	22.56	0.2900	20.70	0.2995	21.64
		MOON	0.6139	30.90	0.3632	26.28	0.3200	23.68	0.3260	24.25	**0.2809**	20.72	0.2966	21.93
		FedDisco	0.6085	30.77	0.3576	26.06	0.3179	23.57	0.3230	24.09	**0.2809**	**20.66**	0.2976	22.05
		FedTPS + FedAvg	**0.6034**	**30.09**	**0.3368**	**24.68**	0.3117	**22.34**	0.3065	**22.14**	0.3047	22.47	**0.2900**	**21.06**
		FedTPS + FedDisco	0.6086	30.47	0.3477	25.66	**0.3070**	22.78	**0.3001**	22.86	0.2872	21.16	0.2926	22.04
	6	Non-FL	0.6637	36.61	0.5366	34.32	0.4429	30.55	0.4519	31.60	0.3761	25.87	0.3351	24.26
		FedAvg	0.6252	32.48	0.3710	26.83	0.3402	24.31	0.3389	24.47	0.3261	23.48	0.3158	22.99
		FedOpt	**0.6213**	32.32	0.3761	26.98	0.3492	24.64	0.3376	24.18	0.3248	23.78	0.3132	23.05
		FedProx	0.6239	32.78	0.3749	26.68	0.3595	24.78	0.3402	24.35	0.3171	22.82	0.3158	22.96
		MOON	0.6257	32.82	0.3786	26.13	0.3522	23.99	0.3418	23.82	**0.3059**	21.61	0.3201	22.53
		FedDisco	0.6237	32.40	0.3702	**25.67**	0.3457	23.94	0.3588	24.73	0.3070	**21.49**	0.3208	22.62
		FedTPS + FedAvg	0.6226	**32.01**	0.3697	26.58	**0.3338**	23.90	0.3286	23.83	0.3363	24.65	0.3132	22.95
		FedTPS + FedDisco	0.6255	32.11	**0.3631**	**25.64**	0.3339	**23.10**	**0.3281**	**22.88**	0.3140	21.67	**0.3130**	**22.18**
	8	Non-FL	0.6875	37.50	0.5199	33.67	0.4911	32.00	0.4708	31.22	0.4217	26.79	0.3759	25.17
		FedAvg	0.6367	33.44	0.3912	25.96	0.3692	24.22	0.3658	23.92	0.3505	23.42	0.3489	23.13
		FedOpt	**0.6333**	33.36	0.3861	25.94	0.3692	24.09	0.3607	24.04	0.3505	23.43	0.3489	23.25
		FedProx	0.6350	33.56	0.3980	26.25	0.3743	24.32	0.3641	24.23	0.3438	22.51	0.3472	23.01
		MOON	0.6364	33.54	0.4058	26.94	0.3784	24.39	0.3711	24.31	0.3332	**22.00**	0.3544	23.33
		FedDisco	0.6354	33.60	0.4021	27.06	0.3777	24.56	0.3680	24.43	**0.3327**	22.05	0.3527	23.35
		FedTPS + FedAvg	0.6384	**33.08**	**0.3793**	25.89	**0.3658**	23.93	**0.3539**	**23.58**	0.3675	24.92	0.3489	23.25
		FedTPS + FedDisco	0.6390	33.49	0.3833	**25.58**	0.3687	**23.85**	0.3618	23.86	0.3427	22.42	**0.3447**	**22.88**
Xi'an	2	Non-FL	0.2389	44.62	0.1719	30.44	0.1716	28.74	0.1516	24.91	0.1387	23.25	0.1300	20.71
		FedAvg	0.2127	33.05	0.1406	20.57	0.1344	20.73	0.1357	21.24	0.1271	19.08	0.1298	19.88
		FedOpt	0.2133	33.44	0.1369	20.13	0.1351	20.40	0.1395	21.51	0.1276	18.81	0.1276	20.18
		FedProx	0.2151	33.31	0.1399	21.56	0.1352	21.29	0.1376	21.63	0.1324	20.35	0.1300	21.05
		MOON	0.2218	31.47	0.1371	20.37	0.1392	20.61	0.1379	20.23	0.1293	18.30	0.1314	19.40
		FedDisco	0.2231	31.43	0.1390	19.97	0.1398	20.43	0.1386	20.70	0.1289	18.72	0.1299	19.41
		FedTPS + FedAvg	**0.2096**	32.09	**0.1256**	18.23	**0.1284**	19.59	**0.1302**	19.96	0.1304	18.60	**0.1240**	19.35
		FedTPS + FedDisco	0.2185	**30.51**	0.1298	**18.20**	0.1301	**18.79**	0.1307	**19.38**	**0.1260**	**17.71**	0.1257	**18.49**
	4	Non-FL	0.2587	38.18	0.2173	34.52	0.2071	32.34	0.1871	28.58	0.1852	26.99	0.1657	25.11
		FedAvg	0.2348	27.00	0.1706	23.08	0.1682	23.58	0.1632	23.38	0.1618	22.15	0.1605	22.34
		FedOpt	0.2354	27.47	0.1698	22.95	0.1682	23.53	0.1632	23.41	0.1627	22.32	0.1597	22.40
		FedProx	0.2362	27.17	0.1736	23.93	0.1693	23.39	0.1632	23.53	0.1618	22.95	0.1594	22.43
		MOON	0.2447	26.36	0.1739	23.06	0.1739	23.03	0.1698	23.09	0.1607	21.36	0.1640	22.04
		FedDisco	0.2439	26.40	0.1778	23.39	0.1728	23.13	0.1678	22.87	0.1612	21.13	0.1630	21.79
		FedTPS + FedAvg	**0.2345**	26.68	**0.1621**	22.06	**0.1613**	22.94	**0.1569**	22.64	0.1709	23.37	**0.1586**	21.81
		FedTPS + FedDisco	0.2407	**26.35**	0.1650	**21.66**	0.1655	**22.12**	0.1641	**22.28**	**0.1604**	**21.16**	0.1595	**21.30**
	6	Non-FL	0.2819	33.64	0.2230	32.15	0.2271	32.56	0.2099	30.39	0.2271	31.77	0.1913	26.23
		FedAvg	0.2535	26.35	0.1963	26.02	0.1975	25.92	0.1930	26.35	**0.1856**	24.40	**0.1823**	24.43
		FedOpt	0.2535	26.52	0.1938	25.96	0.1975	25.69	0.1930	26.82	0.1872	24.25	0.1829	24.54
		FedProx	0.2559	26.69	0.1983	26.85	0.1983	26.08	0.1959	26.49	0.1880	25.05	0.1839	24.83
		MOON	0.2607	26.37	0.2020	**25.20**	0.2055	25.58	0.2005	25.45	0.1868	**23.49**	0.1890	24.14
		FedDisco	0.2619	26.25	0.2021	25.63	0.2020	25.44	0.2024	25.85	0.1875	23.64	0.1890	24.18
		FedTPS + FedAvg	**0.2510**	26.26	**0.1893**	25.79	**0.1913**	**25.39**	**0.1913**	26.34	0.2020	25.91	0.1827	24.60
		FedTPS + FedDisco	0.2591	**25.66**	0.1945	25.29	0.2012	25.42	0.1949	**25.34**	0.1921	23.86	0.1864	**23.66**
	8	Non-FL	0.2935	28.92	0.2403	31.80	0.2480	30.99	0.2359	30.10	0.2529	31.66	0.2238	27.41
		FedAvg	0.2688	27.76	0.2129	28.26	0.2173	28.30	0.2107	27.99	0.2107	26.39	0.2063	26.30
		FedOpt	0.2694	27.79	0.2129	28.20	0.2173	28.01	0.2085	27.92	0.2090	26.52	**0.2052**	26.70
		FedProx	0.2683	28.02	0.2140	28.63	0.2184	28.40	0.2101	27.94	0.2096	26.99	0.2057	26.71
		MOON	0.2758	27.72	0.2204	28.00	0.2226	27.83	0.2136	27.49	0.2090	25.85	0.2106	**25.79**
		FedDisco	0.2762	27.64	0.2181	28.08	0.2242	27.85	0.2135	27.65	**0.2088**	**25.60**	0.2106	25.91
		FedTPS + FedAvg	**0.2677**	27.74	**0.2107**	28.10	**0.2129**	27.68	**0.2074**	27.70	0.2255	28.02	0.2057	26.31
		FedTPS + FedDisco	0.2745	**27.39**	0.2153	**27.37**	0.2188	**27.25**	0.2131	**27.07**	0.2101	26.03	0.2130	26.27

model-dependent and dataset-dependent, with the FedTPS implementations demonstrating the most consistent performance gains across diverse geographic contexts. The FedTPS data augmentation framework can be combined with other FL methods that optimize aggregation protocols or client selection strategies.

With regards to the traffic flow prediction models, performance varies significantly across datasets and FL methods. TAU generally outperforms other state-of-the-art models on the Chengdu dataset, while GATAU shows competitive performance across all three cities. When trained with FedTPS, GATAU demonstrates improved results particularly in the Xi'an dataset across all client configurations, and shows consistent performance advantages in New York. The Graph Attention mechanism can learn cross-region relationships from the entire city, whereas the convolutional operations implemented in TAU restrict spatial dependencies modelling to neighbouring regions. Therefore, data augmentation may benefit GATAU more, as there is more data available to learn cross-region dependencies.

Training TAU under FedTPS shows degraded performance at higher client counts and on the New York dataset compared to standalone FedAvg or FedDisco. Table 4 reveals that New York exhibits substantially lower temporal correlation compared to Chengdu and Xi'an. As TAU's Temporal Attention mechanism relies heavily on learning temporal patterns, the weaker temporal structure in New York and the declining TC at higher client counts suggest that TFDiff-generated synthetic data may not adequately capture these complex temporal characteristics. This results in synthetic data that, while improving spatial representation (evidenced by stable SSI values), provides less value for temporally-oriented architectures.

The effectiveness of FedTPS varies across client configurations, with improvements showing complex dependencies on model architecture, dataset characteristics, and the quality of synthetic data generation. As Table 4 shows, TFDiff's performance (particularly regarding TC) degrades as client numbers increase, suggesting a trade-off between federated learning's data fragmentation and the generative model's ability to capture global spatio-temporal patterns.

D. EFFECTS OF SYNTHETIC PRE-TRAINING ON GLOBAL MODEL TRAINING

Figure 6 details the training loss and evaluation metrics of select traffic prediction models under FL settings, with and without pre-training. With pre-training, the models converge more rapidly than the FL settings without pre-training. However, by global round 60, the models will have generally converged to the same training loss. Hence, while pre-training with synthetic data improves the convergence speed of the global model noticeably, it offers no improved model performance. The random initialisation of model parameters causes the training loss and evaluation metrics at global round 1 to be very high for models without pre-training. However, with pre-training, the starting point of the model is competitive, as can be seen from the low metric scores at global round 1.

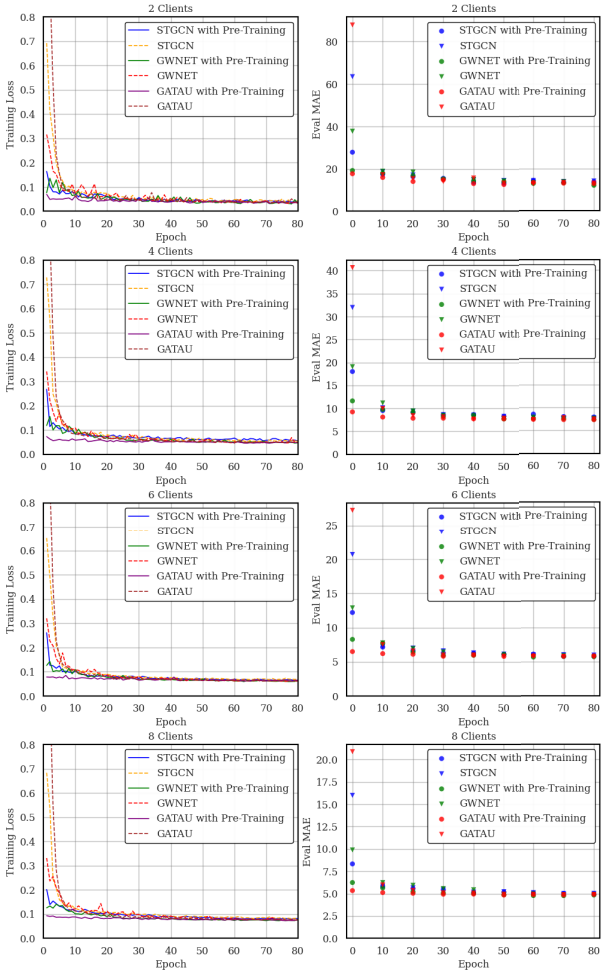


FIGURE 6. Effects of Pre-Training models with Synthetic Data on model training process.

TABLE 6. Computational overhead and communication costs of model training in federated setting, using FedAvg.

	TFDiff	GRU	STGCN	DCRNN	GWNET	TAU	GATAU
Training Time/Epoch (s)	1.41	0.053	0.45	0.90	0.49	0.73	1.03
Training Time/Epoch (s)	-	0.058	0.48	1.10	0.53	0.85	1.15
# Params	12M	66k	216k	302k	446k	13M	12M
Comm. MB/Round	47.52	0.27	0.87	1.79	1.24	50.76	48.40

E. COMMUNICATION COSTS

Table 6 presents the computational and communication overhead for the various different models, trained under a federated setting. The FedTPS framework requires initial training of the TFDiff model, which incurs similar training time per epoch

TABLE 7. Comparison of selected traffic flow prediction models for the city of Chengdu. Models are trained in a federated setting as the ratio of local and global training rounds are varied. FedTPS is implemented here with the FedAvg aggregation mechanism, and compared against standalone FedAvg.

Clients	Model	FL Method	$r_{loc} : r_{\text{glob}}$	nMAE	MAPE
2	STGCN	FedAvg	10	0.1213	23.10
			20	0.1210	23.23
		FedTPS	10	**0.1153**	**20.12**
			20	**0.1145**	**20.58**
	GWNET	FedAvg	10	0.1113	18.88
			20	0.1163	19.70
		FedTPS	10	**0.1085**	**17.95**
			20	**0.1097**	**19.22**
	GATAU	FedAvg	10	0.1192	20.46
			20	0.1124	20.08
		FedTPS	10	**0.1101**	**17.93**
			20	**0.1107**	**18.21**
4	STGCN	FedAvg	10	0.1466	23.76
			20	0.1480	24.58
		FedTPS	10	**0.1410**	**23.68**
			20	**0.1414**	**23.39**
	GWNET	FedAvg	10	0.1384	21.85
			20	0.1383	21.79
		FedTPS	10	**0.1277**	**21.20**
			20	**0.1308**	**21.23**
	GATAU	FedAvg	10	0.1374	20.98
			20	0.1358	21.17
		FedTPS	10	**0.1322**	**20.98**
			20	**0.1329**	**20.39**
6	STGCN	FedAvg	10	0.1633	25.95
			20	0.1637	25.76
		FedTPS	10	**0.1572**	**24.45**
			20	**0.1579**	**24.53**
	GWNET	FedAvg	10	0.1563	23.04
			20	0.1582	23.58
		FedTPS	10	**0.1502**	**22.26**
			20	**0.1525**	**22.24**
	GATAU	FedAvg	10	0.1559	22.33
			20	0.1562	22.51
		FedTPS	10	**0.1516**	**22.04**
			20	**0.1499**	**21.80**
8	STGCN	FedAvg	10	0.1860	28.16
			20	**0.1843**	27.89
		FedTPS	10	**0.1801**	**26.91**
			20	0.1854	**27.73**
	GWNET	FedAvg	10	0.1764	25.01
			20	0.1791	25.52
		FedTPS	10	**0.1710**	**24.57**
			20	**0.1708**	**24.43**
	GATAU	FedAvg	10	0.1772	24.50
			20	0.1775	24.14
		FedTPS	10	**0.1715**	**23.96**
			20	**0.1734**	**23.75**

TABLE 8. Relative improvement (%) of FedTPS+FedAvg over FedAvg baseline on Chengdu dataset across client configurations. Positive values indicate FedTPS provides better performance (lower nMAE).

Clients	GRU	STGCN	DCRNN	GWNET	TAU	GATAU
2	2.01	10.17	9.28	5.68	3.32	4.55
4	2.53	4.63	5.60	5.87	-3.89	5.11
6	2.44	2.33	4.81	3.53	-3.84	4.39
8	1.31	2.54	4.55	3.26	-6.51	2.60
12	0.09	0.72	2.30	1.95	-8.08	-0.05

per client as the TAU and GATAU models. Because FedTPS augments each client's local dataset with synthetic data, the second row shows the increase in training time due to the larger dataset, when the synthetic dataset is composed of 10 days (480 samples). The last row of the table shows that the generative TFDiff model uses less memory per communication round than TAU or GATAU. It should be noted that for cross-silo FL settings, communication and computation bottlenecks are not as prevalent; server-to-client connections represent high-bandwidth data center network connections.

F. EFFECT OF CLIENT ROUNDS AND SERVER ROUNDS

Table 7 illustrates the results of this study, which explores varying the number of local training rounds per global aggregation step. The FedTPS implementation here consistently demonstrates lower error metrics than standalone FedAvg. The decrease in data heterogeneity stemming from the shared synthetic dataset, coupled with the increased information available to each client, means that even in cases where there are greater local training rounds relative to global aggregation rounds, the global model performance is enhanced. This may be because the synthetic data reduces the variance between client updates and promotes more stable and consistent parameter updates. Thus, FedTPS improves the robustness of the FL process against the challenges posed by non-IID data distributions and limited communication rounds.

G. SCALABILITY CONSIDERATIONS

Table 8 shows the relative improvement of FedTPS when implemented with FedAvg aggregation relative to a standalone FedAvg method for the city of Chengdu, as client configurations scale up to 12. As Table 2 shows, mean regional traffic inflow $\bar{q}$ decreases substantially with client count, reflecting the fundamental trade-off between federation granularity and per-client data density in partitions. Very sparse data provides insufficient training signal, constraining maximum feasible client counts. Our $2 - 8$ client experiments shown in Table 5 span the practical range where all datasets maintain adequate density and correspond to realistic urban mobility deployments. As seen in Table 8, FedTPS provides improvements over FedAvg for most models across client counts, demonstrating that synthetic data augmentation provides benefits even in higher client configurations. However, the magnitude

of improvement generally decreases as client count increases. Notably, the TAU model exhibits degradation with FedTPS at higher client counts (4+ clients), consistent with the observations in Section VII-C.

VIII. CONCLUSION

This paper addresses federated learning for traffic flow prediction applications and presents FedTPS, a novel framework that facilitates synthetic data augmentation of client datasets. The work provides theoretical justification through a mixture decomposition framework, demonstrating how federated diffusion models recover global patterns and how synthetic data augmentation reduces data heterogeneity among clients. FedTPS consistently outperforms other FL mechanisms across various client configurations and traffic flow prediction models. Additionally, we introduce GATAU, a novel spatio-temporal prediction model that combines Graph Attention and Temporal Attention mechanisms, demonstrating improved performance when trained with synthetic data augmentation.

The framework is evaluated using large-scale datasets spanning three diverse cities: ride-sharing trajectory data from Chengdu and Xi'an, and taxi origin-destination data from New York. This multi-modal evaluation demonstrates FedTPS's broad applicability across different urban contexts and data collection paradigms.

Future work could explore FedTPS applications in highly non-IID data settings and investigate adaptive synthetic data generation based on client-specific data characteristics. Additionally, while our framework operates on aggregated traffic flow data rather than individual trajectories, formal quantitative privacy assessment remains an important gap. Future work should evaluate vulnerability to trajectory reconstruction attacks to provide rigorous privacy guarantees.

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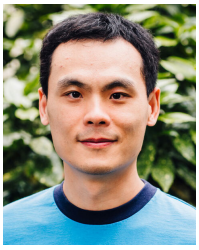
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